



Theroy of Computation

Theory of Computation (Gujarat Technological University)

Unit → 1Review of Mathematical Theory⇒ Definition :-

- It is a branch of computer science that deals with whether and how efficiently Problems can be solved on a model of computation using an algorithm.

⇒ Different names of TOC :-

- Theory of Computation
- Theory of computers
- Theory of computing.

⇒ Different Machines & their languages.

<u>Machines</u>	<u>Languages</u>
- Finite Automaton	Regular languages
- Push down Automaton	Context free languages
- Turing Machine	Recursively Enumerable languages

✓ Purpose: Develop formal mathematical models of computation that reflect real world computers. Date: _____

⇒ Application of TOC (4 M)

IMP

- Implementation of computer science is based on TOC.

1) The TOC aims at understanding the nature of computation & specifically the inherent possibilities & limitation of efficient computations.

2) It concerned with finding the most efficient method for solving specific problems.

ex:- Multiplying num. can be done more efficiently than via the simple method learned in elementary schools.

3) It is used in designing and checking the behavior of digital circuits.

4) In building compiler component, like "lexical analyzer" of typical compiler.

5) In building software for large bodies of text, such as collection of web-pages, to find occurrences of words.

Phrases or Other Patterns.

software for verifying systems of all types that have finite no. of distinct states. such as communication Protocol or secure exchange of information Protocol.

Sets :-

A set is a collections of elements without any structure or order.

In other ~~ex~~ words set may be defined as a collection of objects.

A set is determine by its elements. An elements is known as a member of sets.

Note: Notes zero x copy attached.
(set)

* Sets :-



A set is a collection of elements w/h any structure or order.

In other words sets may be defined as a collection of objects.

→ A set is determined by its elements

→ An element is known as a member of set.

→ An easy way to describe or specify a finite set is to list all its elements for ex.

$$A = \{11, 12, 21, 22\}$$

→ Order of elements in which we write is irrelevant.

i.e. order is no matter u can write your set like this also

$$A = \{11, 22, 21, 12\}$$

→ Writing on elements more than once, does not change the set.

The sets $\{11, 21, 22, 11, 12, 21\}$ and $\{11, 21, 22, 12\}$ are the same.

⇒ Even if a set is infinite, it may be possible to start listing the elements in a way that makes it clear what they are.

for example,

$$B = \{3, 5, 7, 9, \dots\}$$

describes set of all odd integers value.

⇒ A set can also be described by listing the properties that its member must satisfy.

$$C = \{x \mid 1 \leq x \leq 2, \text{ and } x \text{ is a real no.}\}$$

represents set of real no. betⁿ 1 & 2.

$$\{11, 21, 22, 11\} = A$$

* Examples of Set

(i) $A = \{1, 3, 5, 7, 9, \dots\}$, a set of odd no.

(ii) $A_2 = \{a, e, i, o, u\}$, a set of vowels

(iii) $A_3 = \{1, 2, 3, 5, 7, 11, 13, \dots\}$ set of all prime no.

(iv) $A_4 = \{aa, aaaa, aaaaaa, \dots\}$
set of strings of a's with even occurrence

The method by which above sets are represented known as Roster or tabulation method.

→ In this method the elements of the sets are listed within brackets & separated by comma.

⇒ Another method to represent a set is known as set-builder method,

→ In this method the elements of set are represented by concluded properties.

Ex. $A = \{\omega \mid \text{Properties of } \omega\}$

Ex. $A = \{x \mid x \text{ is an odd integer } > 1\}$

$A_1 = \{x \in \mathbb{N} \mid x \cdot 2 = 1 \text{ \& } N \text{ is natural no.}\}$

$A_2 = \{x \in E \mid x \text{ is a vowel from English alphabet}\}$

$A_3 = \{x \in P \mid x \text{ is a prime no.}\}$

$A_4 = \{x \in \mathbb{Q}^n \mid n \geq 2 \text{ \& } n \% 2 = 0\}$

* Cardinality $\rightarrow$

The number of distinct elements in a set is called the cardinality of a set.

for ex.

$A = \{0, 1, 2\}$

elements in $A = 3$ So,

Cardinality of set A is 3

f.c. $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\} = A$

* Types of sets :-

(i) Equivalent sets :-

Two finite sets A & B will be called equivalent sets if and only if they have the same cardinality.

for ex. $A = \{0, 1, 2\}$, $B = \{a, b, c\}$

(ii) Equal set / Equality of set :-

Two sets A and B will be said to be equal set if and only if they have the same elements.

ex. A & B represented as

$$A = \{a, e, i, o\}, B = \{a, e, i, o\}$$

or

$$A = \{1, 2, 3\} = \{3, 2, 1\}$$

or

$\{1, 2, 3\} = \{3, 2, 1, 1\}$ all are equal set

(iii) Empty set :-

An empty set is defined as a set of no elements. It is also called as "**NULL SET**".

→ It is denoted by ϕ

→ An empty set is a sub-set of all sets.

(iv) Singleton Sets :-

A set containing only one (single) element is called a Singleton set.

for $A_1 = \{0\}$, $A_2 = \{1\}$, $A_3 = \{x\}$

(v) Sub-Set :-

If every elements in a set A is also an elements of set B, then A is called a sub-set of B. It is written as

$$A \subseteq B \text{ or } B \supseteq A$$

(vi) Super-Set :-

When A is a subset of B (i.e. $A \subseteq B$) - then B is a super set of A which is written as $B \supset A$

Ex Let us consider sets A and B defined in the following ways:

1. $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4, 5\}$ then $A \subseteq B$

2. $A = \{x \mid x \text{ is a vowel in Eng.}\}$,

$B = \{x \mid x \text{ is an English alphabet}\}$
then $A \subseteq B$

3. $A = \{x, y, z\}$

$$B = \{\gamma, \beta, \alpha\}$$

then $A \subseteq B$ and $B \subseteq A$

vii * proper subsets $\rightarrow$

If $A \subseteq B$ then it is still possible that $A = B$.
 $\rightarrow$ When $A \subseteq B$ but $A \neq B$, we say A is a proper subset of B .
 we write it $A \subset B$ when A is a proper subset of B .

for example, $A = \{1, 2\}$ $A \subset B$
 $B = \{1, 2, 3\}$ $B \subset C$
 $C = \{1, 3, 2\}$ $A \neq B$

then both A and B are subsets of C but A is a proper subset of C

whereas C is not a proper subset of B .
 Since $C = B$

viii * Finite set $\rightarrow$

A set is said to be finite if it contains exactly n distinct elements where n denotes some non-negative integer, o.w., the set is said

Ex. $A = \{1, 1, 2, 2, 2\} \rightarrow$ finite set

$A = \{1, 3, 5, 7, 9, \dots\} \rightarrow$ infinite set

Power Set :- $\rightarrow$

The power set of a set S is denoted by 2^S . If we have a set Q defined as $Q = \{q_1, q_2\}$, then the power set of Q has elements

$$2^2 = 2^2 \quad (\emptyset, \{q_1\}, \{q_2\}, \{q_1, q_2\})$$

It is a set of all subsets of a given set.

Universal Set :-

A set U is called universal set if U is the superset of all the sets which are under our consideration.

$$\text{Ex. } A = \{1, 3, 5, 7, 9\}$$

$$B = \{2, 4, 6, 8, 10\}$$

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A \subseteq U \quad \& \quad B \subseteq U$$

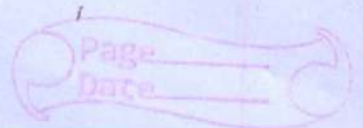
$\therefore U$ is the Universal Set of A and B .

Complement Set :- $\rightarrow$

The complement of a set A is the set A' of everything that is not an element of A . This makes sense only in the context of some "universal set" U containing

all the elements,

i.e. $A' = \{x \in U \mid x \notin A\}$



* Disjoint Set :-

Two sets A and B are said to be disjoint set if

$$A \cap B = \{\emptyset\}$$

Ex. $A = \{a, b, c, d\}$ $B = \{p, q, r, s\}$

$$A \cap B = \{\emptyset\}$$

$\therefore$ A & B are disjoint set.

* Theorem

1) For an arbitrary set A, $A \subseteq U$

2) " " " " " " , $\emptyset \subseteq A$

3) " " " " " " , $A \subseteq A$

4) " " " " " " , the number of subsets of A is $2^{|A|}$

Ex. List the members of following sets:

(a) $\{x \mid x \text{ is a real no. such that } x^2 = 4\}$

(b) $\{x \mid x \text{ is an integer such that } x^2 = 2\}$

Ans (a) $\{-2, 2\}$, (b) $\emptyset$

2

Determine whether each of the following pairs of sets is equal or not?

(a) $\{1, 2, 3, 1, 1, 2\}$, $\{2, 3, 1\}$

equal

Yes

(b) $\{\{1\}\}$, $\{1, \{1\}\}$

No

(c) $\emptyset$, $\{\emptyset\}$

No.

3 For each of the following sets, determine whether 1 is an elements of that set.

(a) $\{x \in \mathbb{R} \mid x \text{ is an integer greater than } 1\}$

Ans. No.

(b) $\{x \in \mathbb{R} \mid x \text{ is the square of an integer}\}$

Yes

(c) $\{1, 2, \{1\}\}$

Yes

(d) $\{\{1\}, \{\{1\}\}\}$

No.

(e) $\{\{1, 2\}, \{1, \{1\}\}\}$

No

(f) $\{\{\{1\}\}\}$

No.

Ex

Suppose that A , B and C are sets such that $A \subset B$ & $B \subset C$. Show that $A \subset C$.

Let x be an arbitrary elements of A

i.e. $x \in A$

we have to prove $x \in C$

Now $A \subset B$

i.e. $x \in B$

Now $B \subset C$

i.e. $x \in C$

Since $x \in A$ implies $x \in C$

it follows that $A \subset C$

* find the power set of each of the following set.

(a) $\{a\}$ (b) $\{a, \{a\}\}$ (c) $\{\emptyset, \{\emptyset\}\}$

~~Ans~~
(a) $\{\emptyset, \{a\}\}$

(b) $\{\emptyset, \{a\}, \{\{a\}\}, \{a, \{a\}\}\}$

(c) $\{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$

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* operation on sets:

(i) Union & intersection of sets: →

→ The union of two sets A and B, denoted by $A \cup B$, is the set defined by the set of elements which belong to A or B.

for example :-

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

→ The intersection of two sets A & B is denoted by $A \cap B$. It is defined by the set of elements which belong to both A and B.

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

Ex $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 6, 8, 10, 12\}$

$$A \cup B = \{1, 2, 3, 4, 5, 6, 8, 10, 12\}$$

$$A \cap B = \{2, 4, 6\}$$

$$(2) \quad A = \{x \mid x \text{ is an english alphabet}\}$$

$$B = \{x \mid x \text{ is vowel in english alphabet}\}$$

$$\begin{aligned} \therefore A \cup B &= \{x \mid x \text{ is an english alphabet}\} \\ &= \{a, b, c, d, e, f, \dots, z\} \\ &= A \end{aligned}$$

$$\begin{aligned} A \cap B &= \{x \mid x \text{ is vowel in english alphabet}\} \\ &= \{a, e, i, o, u\} \\ &= B. \end{aligned}$$

$$(3) \quad A = \{a, b, c\}, \quad B = \{0, 1, 2, 3\}, \quad C = \{A, B\}$$

$$A \cup B \cup C = \{a, b, c, 0, 1, 2, 3, A, B\}$$

$$A \cap B \cap C = \{\phi\}$$

* Cartesian product / cross-product :-

If A and B are two sets, the cross-product of A & B written $A \times B$ is the set of all pairs wherein the first element is a member of A & the 2nd element is a member of B .

$$\text{If } A = \{1, 2\} \quad B = \{x, y, z\}$$

$$A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}$$

* Complement $\rightarrow$

The absolute complement or simply complement of a set A , denoted by A' or $\bar{A}$, is the set of elements which belong to U but which do not belong to A ,

i.e. $A' = \{x | x \in U, x \notin A\}$

Ex:

$$A = \{2, 4, 6, 8, 10\}$$

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A' = \{1, 3, 5, 7, 9\}$$

* Set Difference $\rightarrow$

The set difference of a set A with respect to set B denoted by $A - B$ is the set of all elements that are in A but not in B .

$$A - B = \{x | x \in A \text{ \& } x \notin B\}$$

Ex:

$$A = \{a, c, d, g, h, i\}$$

$$B = \{a, b, c, e, f, g, i\}$$

then

$$A - B = \{d, h\}$$

$$B - A = \{b, e, f\}$$

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* Symmetric diff. :-

The symmetric diff. of two sets A & B is defined as

$$\underline{A \oplus B} \text{ or } \underline{A \Delta B} = \{ (A-B) \cup (B-A) \}$$

$$A = \{ 1, 3, 5, 7, 10 \}$$

$$B = \{ 1, 2, 4, 5, 6, 8, 9, 10 \}$$

$$A - B = \{ 3, 7 \}$$

$$B - A = \{ 2, 4, 6, 8, 9 \}$$

$$\therefore A \Delta B = (A - B) \cup (B - A)$$

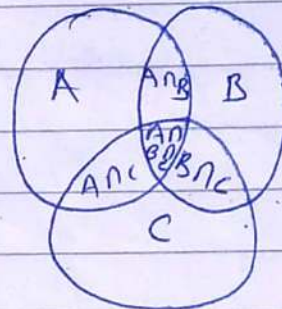
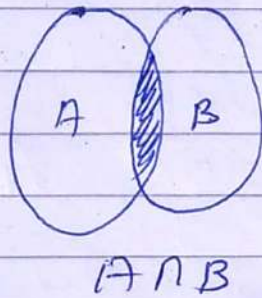
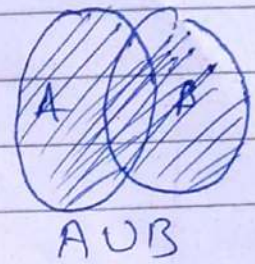
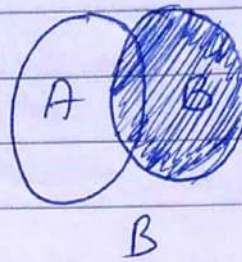
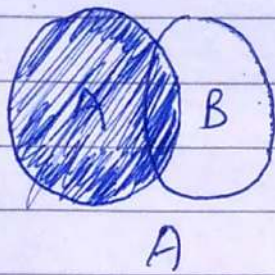
$$= \{ 3, 7 \} \cup \{ 2, 4, 6, 8, 9 \}$$

$$= \{ 2, 3, 4, 6, 7, 8, 9 \}$$

* Representation of set operations :-

By using a Venn diagram, set operations can be represented easily.

In a Venn diagram, a universal set is represented by a rectangle and the remaining sets are represented by circles or ellipses.

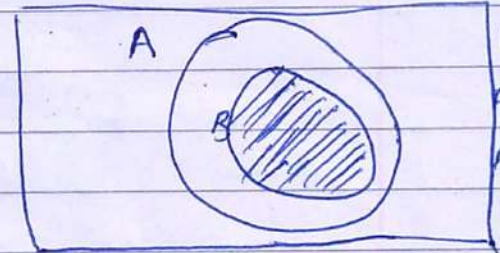
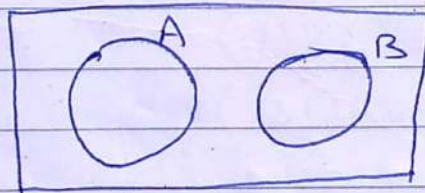


Disjoint

$$A \cap B = \emptyset$$

$$A \cup B =$$

$$n(A) + n(B)$$



Subset

$$B \subseteq A$$

$$A \cap B = B$$

$$A \cup B = A$$

* List of some standard set identities.
In all rules a, b, c represent arbitrary sets. U represent universe set & ϕ is the empty set.

① The commutative laws:-

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

② The associative laws:-

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

③ The distributive laws:-

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

④ The idempotent laws:-

$$A \cup A = A$$

$$A \cap A = A$$

⑤ The absorptive laws:-

$$A \cup (A \cap B) = A$$

$$A \cap (A \cup B) = A$$

(6) The De Morgan laws:-

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

(7) laws involving complements:-

$$(A')' = A$$

$$A \cap A' = \emptyset$$

$$A \cup A' = U$$

(8) laws involving the empty set:-

$$A \cup \emptyset = A$$

$$A \cap \emptyset = \emptyset$$

(9) laws involving universal set:-

$$A \cup U = U$$

$$A \cap U = A$$

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proof of De Morgan laws:-

$$(A \cap B)' = A' \cup B'$$

To prove this that two sets are equal we will show that each of the two sets is a subset of the other.

So,

$$(A \cap B)' \subseteq A' \cup B'$$

Let

$$x \in (A \cap B)'$$

$$\text{i.e. } x \notin A \cap B$$

$$\text{i.e. } x \notin A \text{ or } x \notin B$$

$$\text{i.e. } x \in A' \text{ or } x \in B'$$

$$\therefore x \in A' \cup B'$$

$$\text{So, } (A \cap B)' \subseteq A' \cup B'$$

Now

$$A' \cup B' \subseteq (A \cap B)'$$

$$x \in A' \cup B'$$

$$\therefore x \in A' \text{ or } x \in B'$$

$$\therefore x \notin A \text{ or } x \notin B$$

$$\therefore x \notin (A \cap B) \quad x \notin A \cap B$$

$$\therefore x \in (A \cap B)' \quad x \in (A \cap B)'$$

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⇒ Logical statements

- which one and only one of the terms true or false.

ex:- "Five is odd" → True

" $5 \in \{1, 2, 3\}$ " → False

- used in mathematics, circuits etc...

⇒ Logical Propositions :-

Same def :-

⇒ Free variables :-

The answer of proposition statement depend on value of some variable then such variable are called free variables.

ex:- $a^2 + b^2 = 4$

(Here ans. 4 depend on the two variables then a & b are the free variable).

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⇒ Logical ~~connections~~ connectives :-

- 1) Conjunction
- 2) Disjunction
- 3) Negation

→ Conjunction :- (AND)

$P \wedge Q$. It is denoted by $\wedge$.

It is read as P and Q.

[Note: and operation is performed]

[Multiplication]

Truth Table

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

ex:- P : Google is website (T)
= Q : Google is search engine (T)

$P \wedge Q$ = Google is website and search engine.

$T \wedge T = T$ (value of statement)

Date : _____

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[Note: and operation is performed]

[Multiplication]

Truth Table

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
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ex:- P : Google is website (T)
= Q : Google is search engine (T)

$P \wedge Q$ = Google is website and search engine.

$T \wedge T = T$ (value of statement)

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2) Disjunction :- (OR)

- denoted by $P \vee Q$.
- read as a P or Q

[Note: OR operation is performed]

⇒ Truth table :

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

ex:- P : Google is website (T)
 Q : Google is search engine (T)

 $P \vee Q$: Google is website or search engine

T ∨ T = T (value of statement)

3) Negation :- (NOT)

- It is denoted by "¬" or "~".
- read as "NOT"

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⇒ Truth Table :

P	$\sim P$
T	F
F	T

ex:- P : Google is website. $\sim P$: Google is not a website.⇒ Other logical connectives :-1) Conditional :-

- denoted by $P \Rightarrow Q$ or $P \rightarrow Q$.
- Read as a → if P then Q.

P	Q	$P \Rightarrow Q$	$Q \Rightarrow P$	$\sim P$	$\sim P \vee Q$
T	T	T	T	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Note:- $P \Rightarrow Q = \sim P \vee Q$ Proved.

ex:- P : If there is raining
 Q : roads are wet.

$P \Rightarrow Q$: if there is raining then roads are wet.

2) Biconditional (Bidirectional) :-

- denoted by $P \Leftrightarrow Q$ or $P \leftrightarrow Q$
- read as " P if and only if Q "

Note: $P \Leftrightarrow Q = (P \Rightarrow Q) \wedge (Q \Rightarrow P)$

P	Q	$P \Leftrightarrow Q$	$P \Rightarrow Q$ $Q \Rightarrow P$		$(1) \wedge (2)$
			(1)	(2)	
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

$\Rightarrow$ Hence Proved.

3) Tautology :-

- An expression involving logical variables that is true in all cases is called tautology.

- It means answer is always true.

Note: $P \vee \sim P = T$ [$\because P=1 \sim P=0$
 $1+0=1=T$]

P	$\sim P$	$P \vee \sim P$
T	F	T
F	T	T

ex:- (i) $\sim (P \wedge Q) \Leftrightarrow (\sim P) \vee (\sim Q)$

(ii) $P \vee \sim P$

$\vee$ (iii) $(P \wedge Q) \rightarrow P$

(iv) $Q \rightarrow (P \vee Q)$

(v) $(P \vee Q) \Leftrightarrow (Q \vee P)$

~~(vi) $P \rightarrow (P \wedge Q)$~~

solve:-

$$P \vee Q \vee \sim S \Leftrightarrow (\sim P \wedge \sim Q \wedge \sim S) \rightarrow S$$

LHS

RHS

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ex: (vii) $P \rightarrow (\sim P \vee \sim Q)$ Tautology or not?

P	Q	$\sim P$	$\sim Q$	$\sim P \vee \sim Q$	$P \rightarrow (\sim P \vee \sim Q)$
T	T	F	F	F	F
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	T	T

Not tautology

4> contradiction:-

- Opposite of tautology which is false in every case.
- It means answer is always false.

ex: $P \wedge \sim P = F$

P	$\sim P$	$P \wedge \sim P$
T	F	F
F	T	F

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ex: $P \vee (P \Rightarrow Q)$ is this statement is tautology? then justify your answer

P	Q	$P \Rightarrow Q$	$P \vee (P \Rightarrow Q)$
T	T	T	T
T	F	F	T
F	T	T	T
F	F	T	T

all answers true.

 $\therefore$ tautology. Proved.ex: Proof that $P \rightarrow Q \equiv \sim P \vee Q$

P	Q	$\sim P$	$P \rightarrow Q$	$\sim P \vee Q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

(i) (ii)

(i) & (ii) are similar.

Hence Proved.

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⇒ Logical Implication & Equivalence.

- $P \Rightarrow Q$: If Q is true in each case in which P is true.

Note:- logical connective $\rightarrow$
and

relation $\Rightarrow$ of logical
equivalence are similar.

- ✓ If P & Q have the same truth table in each case, we say P & Q are logically equivalent.

$$P \Leftrightarrow Q \iff P \Rightarrow Q \text{ \& } Q \Rightarrow P$$

⇒ Logical Quantifiers & Quantified Statement

$$x^2 < 4$$

In this statement x is missing.

- If we attaching the phrase,

there exists an x such that
 x is no longer a free variables.

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- so we say that the statement

"there exists an x such that $x^2 < 4$ "
is a quantified statement.

we write it $\exists x (x^2 < 4)$ [there exists
atleast one x]

⇒ another notation $\forall \Rightarrow$ for all

$$\text{ex: } \forall x (x^2 < 4)$$

• Advantage:-

- ① force you to specify
- ② easy to understand the scope of each quantifier.

• Note:-

- ① $\neg \forall x (P(x))$ is same as $\exists x (\neg P(x))$
- ② $\neg \exists x (P(x))$ is same as $\forall x (\neg P(x))$

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⇒ Functions :-

IMP:

A function is an object that sets-up an input-output relationship.

- A function takes an input & produces an output.

- If f is a function & its output value is b when the input value is a , then we write,

$$f(a) = b$$

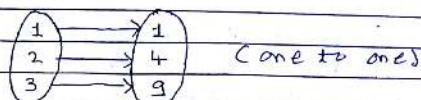
- A function also called a mapping, and if $f(a) = b$, we say that f maps a to b .

⇒ $f: A \rightarrow B$
 $\downarrow \quad \downarrow$
 (domain) (co-domain)

⇒ $f: \mathbb{R} \rightarrow \mathbb{R}$
 $\downarrow \quad \downarrow$
 (domain) (co-domain)

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⇒ $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$

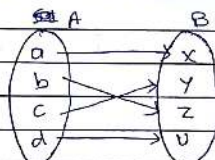


⇒ 3 Main type of functions

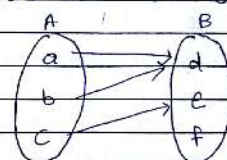
- (i) one to one
- (ii) on to
- (iii) Bijection

(i) one to one function (injective) (injection)

- A function $f: A \rightarrow B$ is said to be one to one, if and only if for all distinct elements of set A map to all distinct elements of set B .



one to one mapping



not one to one mapping

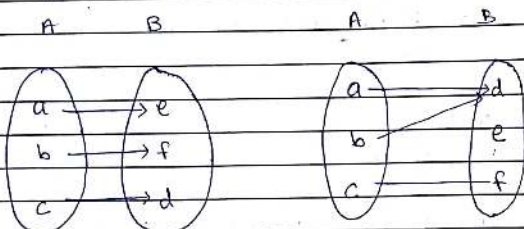
Note:

$f: A \rightarrow B$ is said to be one to one if no two elements of A have an image in B .

✓

(ii) Onto function (surjective) (Surjection)

- function $f: A \rightarrow B$ is said to be onto if every elements of B is mapped by some elements in A .



(onto)

(not onto)

(iii) Bijection

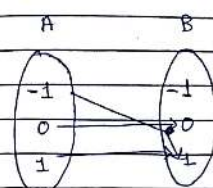
- If the function is one to one and onto then it is called as Bijection.

ex:- $f: \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = x^2$.
check whether this function is one to one or onto?

$\Rightarrow f: \mathbb{R} \rightarrow \mathbb{R}$

$\mathbb{R}: \{ \dots, -2, -1, 0, 1, 2, \dots \}$

domain co-domain



[$\therefore$ It is not onto because one is remaining in co-domain]

$$f(x_1) = f(x_2)$$

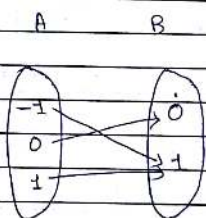
$$f(-1) = f(+1)$$

Hence $-1 \neq 1$ (hence it is not one to one)

ex:- $f: \mathbb{R} \rightarrow \mathbb{R}^+$ where $f(x) = x^2$.

$$\Rightarrow \mathbb{R} = \{ \dots, -2, -1, 0, 1, 2, \dots \}$$

$$\mathbb{R}^+ = \{ 0, 1, 2, \dots \}$$



$$f(-1) = 1$$

$$f(0) = 0$$

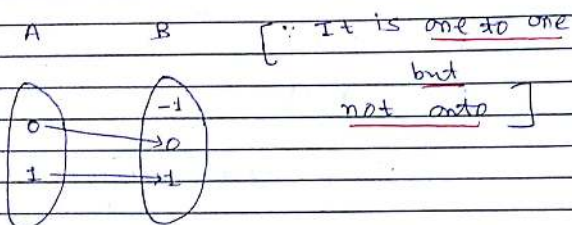
$$f(1) = 1$$

[$\therefore$ Not one to one but onto $\mathbb{R}^+$]

ex:- $f: \mathbb{R}^+ \rightarrow \mathbb{R}, f(x) = x^2$.

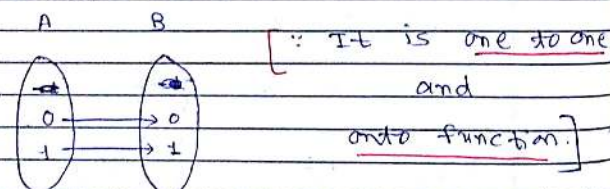
$\Rightarrow \mathbb{R}^+ = \{0, 1, 2, \dots\}$

$\mathbb{R} = \{\dots, -2, -1, 0, 1, 2, \dots\}$



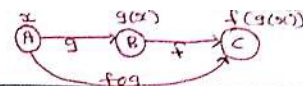
ex:- $f: \mathbb{R}^+ \rightarrow \mathbb{R}^+, f(x) = x^2$

$\Rightarrow \mathbb{R}^+ = \{0, 1, 2, \dots\}$



$\left[\because \text{It is also called Bijection.} \right]$

ex:- $f: \mathbb{N} \rightarrow \mathbb{N}, f(x) = 2x$ Bijection or Not?



$\Rightarrow$ Composition function :-

$f: A \rightarrow B$ & $g: B \rightarrow C$ then for any $x \in A$, $f(x)$ is in the domain of g and it is therefore,

$h: A \rightarrow C, h(x) = g(f(x))$ is called composition of g & f and is written as ;

$h = g \circ f$

f^m :- $\begin{matrix} f \circ f = f(f(x)) & f \circ g = f(g(x)) \\ g \circ f = g(f(x)) & g \circ g = g(g(x)) \end{matrix}$

ex:- If $f(x) = x^2, g(x) = x^3$ then find $f \circ g, g \circ f, g \circ g, f \circ f$.

$\Rightarrow f \circ f = f(f(x)) = f(x^2) = (x^2)^2 = x^4$

$\bullet g \circ f = g(f(x)) = g(x^2) = (x^2)^3 = x^6$

$\bullet g \circ g = g(g(x)) = g(x^3) = (x^3)^3 = x^9$

$\bullet f \circ g = f(g(x)) = f(x^3) = (x^3)^2 = x^6$

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ex: let f & g are function from $\mathbb{R} \rightarrow \mathbb{R}$ defined as,

$$f(x) = x^2 + 9 \quad g(x) = 4x + 5$$

find $f \circ g$, $g \circ f$, are they equal?

$$\Rightarrow \bullet f \circ g = f(g(x)) = f(4x + 5)$$

$$\text{Now, } f(x) = x^2 + 9$$

$$= (4x + 5)^2 + 9$$

$$= 16x^2 + 40x + 25 + 9$$

$$= 16x^2 + 40x + 34 \quad \text{--- (i)}$$

$$\bullet g \circ f = g(f(x)) = g(x^2 + 9)$$

$$\text{Now, } g(x) = 4x + 5$$

$$= 4(x^2 + 9) + 5$$

$$= 4x^2 + 36 + 5$$

$$= 4x^2 + 41 \quad \text{--- (ii)}$$

$\therefore$ from (i) & (ii), so we can prove that

$$f \circ g \neq g \circ f.$$

ex: let f & g are f^n from $\mathbb{R}$ to $\mathbb{R}$ defined as $f(x) = x^2 + 3$, $g(x) = 4x - 5$ find $f \circ g$ & $g \circ f$, are they equal? Ans: Not equal.

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Note: If $f: A \rightarrow B$ & $g: B \rightarrow C$ are one to one $g \circ f$ is also one to one.

$\rightarrow$ If f & g are onto, $g \circ f$ is also onto.

$\rightarrow$ If f & g are Bijective, $g \circ f$ is also Bijective.

$\Rightarrow$ Inverse function :-

- A function $f: A \rightarrow B$ with another function $g: B \rightarrow A$ is said to be Inverse f^n if the function $g: B \rightarrow A$ associates each element $b \in B$ to a unique $a \in A$, such that

$f(a) = b$ is called the Inverse of $f: A \rightarrow B$.

OR

- Let f be a one to one function with domain a and range b . then its inverse function f^{-1} has domain b and range a .

- It is defined by

$$f^{-1}(y) = x \Leftrightarrow f(x) = y \text{ for any } y \text{ in } b.$$

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⇒ for every $x \in A$, $f^{-1}(f(x)) = x$
for every $y \in B$, $f(f^{-1}(y)) = y$.

ex:- $f(x) = x^3$ find $f^{-1}(x)$.

⇒ (1) Replace $f(x)$ by $y \Rightarrow y = x^3$

(2) solve for $x \Rightarrow x = \sqrt[3]{y}$

(3) replace x by $f^{-1}(x)$ and y by x .

$$f^{-1}(x) = \sqrt[3]{x}$$

ex:- find the inverse of function $f(x) = \frac{x+1}{x}$

⇒ (1) Replace $f(x)$ by $y \Rightarrow y = \frac{x+1}{x}$

(2) solve for $x \Rightarrow xy = x+1$
 $\therefore xy - x = 1$
 $\therefore x(y-1) = 1$

$$\therefore x = \frac{1}{y-1}$$

(3) Replace x by $f^{-1}(x)$ & y by x

$$\therefore f^{-1}(x) = \frac{1}{x-1}$$

Domain of R : set of all first element.
Range of R : set of all second element.
Date: _____

⇒ Relation :-

- A mathematical relation is a way of making more precise the idea of a relationship between object.

✓ - A relation on a set A is a subset of $A \times A$.

⇒ Suppose R is a relation from A to B , then R is a set of ordered pair, where each first element comes from A and each second elements comes from B .

✓ $(a, b) \in R \Rightarrow$ i.e. a is R -related to b aRb

✓ $(a, b) \notin R \Rightarrow a$ is non-related to b .

eg:- Two sets N_1 & N_2 of natural no. If we defined R as relation betⁿ them such that N_1 is cube of N_2 then

$1R1, 2R8, 3R27, \dots$

so these relations can also written as

$$R = \{(1,1), (2,8), (3,27), \dots\}$$

$$= \{(a,b) \mid a, b \in N, \text{ or } N_2 \text{ \& } b = a^3\}$$

ex:-

If $A = \{1, 2, 3, 4, 5\}$ & $B = \{6, 7, 8, 9, 10\}$ and
 suppose $R = \{(1, 7), (2, 8), (5, 9), (5, 10)\}$
 then;

domain $R = \{1, 2, 5\}$

Range $R = \{7, 8, 9, 10\}$

⇒

Inverse of Relation :-

- If R is a relation from a set A to set B , then the inverse of R , denoted by R^{-1} .
- Means the relation from B to A which consists of those ordered pairs which, when reversed belong to R .

eg:- $R^{-1} = \{(b, a) \mid (a, b) \in R\}$

let $[\because R: A \rightarrow B \Rightarrow R^{-1}: B \rightarrow A]$

$A = \{a, b, c, d\}$ $B = \{1, 2, 3, 4\}$

$R = \{(a, 1), (a, 3), (b, 3), (c, 4)\}$

so, $R^{-1} = \{(1, a), (3, a), (3, b), (4, c)\}$

* Types / Properties of Relation: - A (E)

- (1) Reflexive ✓
- (2) Symmetric ✓
- (3) Antisymmetric ✓
- (4) Transitive ✓

(1) Reflexive :-

A relation R on a set A is reflexive if aRa for every $a \in A$.

$$\boxed{\forall a \in A \Rightarrow aRa}$$

✓
(P)

(2) Symmetric Relation :-

A relation R on a set A is said to be symmetric if whenever aRb then bRa . i.e. whenever $(a,b) \in R$ then (b,a) is also $\in R$.

$$\boxed{\text{if } aRb \text{ then } bRa}$$

Ex.

- (i) Let l_1 & l_2 two line parallel to each other i.e. $l_1 \parallel l_2$ & $l_2 \parallel l_1$ then relation "parallel" is said to be symmetric.

$$R_1 = \{(1,3), (3,1), (2,4), (4,2)\} \text{ etc.}$$

R_1 is Symmetric Relation.

③ Anti-symmetric Relation :-

Relation set R is said to Anti-symmetric if there exists $(a,b) \in A$ such that $(a,b) \in R$ but $(b,a) \notin R$.

if aRb and but not bRa

Ex. Let $A = \{1, 2, 3\}$
 $R = \{(1,3), (3,1), (2,4)\}$

We have $(2,4) \in R$ But $(4,2) \notin R$ so
 Relation set R is Anti-symmetric.

④ Transitive Relation :-

A relation R on a set A is said to be transitive if whenever aRb and bRc then aRc , i.e. whenever

$(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$.

if aRb and bRc then aRc

* Equivalence Relation :-

A binary relation R on a set A will be called an equivalence relation if and only if

- ① R is reflexive relation
- ② R is symmetric relation
- ③ R is transitive

* Equivalence Class :-

For equivalence relation R on set A , the set A , the set of elements of A that are related to an element, say a of A is called the equivalence class of element a and it is denoted by $[a]$.

Let R_1 & R_2 be equivalence relations then $R_1 \cap R_2$ is equivalence relation. But $R_1 \cup R_2$ is not necessarily an equivalence relation.

Exercise:-

- (1) In each case, a relation on the set $\{1, 2, 3\}$ is given at the three properties, reflexivity, symmetric & transitivity. Determine on the relation has, give reason

$\rightarrow (a) R = \{(1, 3), (3, 1), (2, 2)\}$

- (i) Reflexive :- No because for 2, 3
Don't have $(2, 2)$ & $(3, 3)$ relation.

- (2) Symmetric :- Yes,
each pair having aRb & bRa relation

- (3) Transitive :- No
 aRb & $bRc \Rightarrow aRc$
 $1R3$ $3R1$ but not $(1, 1)$

$\rightarrow (b) R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$ $A = \{1, 2, 3\}$

- (1) Reflexive :- Yes
 $\forall a \in R$ we have aRa relation

- (2) Symmetric :- No
for $(1, 2)$ we don't have $(2, 1)$ relation.

- (3) Transitive :- Yes
 $(1, 1) (1, 2) \Rightarrow (1, 2)$
 ~~$(1, 1) (1, 1) \Rightarrow (1, 1)$~~ but not

$\rightarrow (c) R = \emptyset$

- (i) reflexive :- No,
 $\forall a \in A$ aRa Not present.

- (2) Symmetric :- Yes
because if aRb then bRa
But, aRb is not present in R then no need to check.

- (3) Transitive :- Yes
if aRb & $bRc \Rightarrow aRc$
but no aRb & bRc so no need to check aRc .

$\rightarrow (d) R = \{(1, 1), (2, 2), (2, 3), (3, 2), (4, 4), (4, 1)\}$

- (1) Reflexive :- No $3R3$ not exist

- (2) Symmetric :- No $4R2$ but no $2R4$

- (3) Transitive :- No $4R2$ & $2R3$ but no $4R3$.

$\rightarrow A = \{1, 2, 3\}$ relation to R

Relation	Reflexive	S.	T.
(1) $\{(1, 1), (2, 2), (3, 3)\}$	Y	Y	Y
(2) $\{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3), (3, 2), (1, 3)\}$	Y	Y	Y
(3) $\{(1, 1), (2, 2), (3, 3), (1, 2)\}$	Y	N	Y
(4) $\{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3)\}$	Y	N	Y
(5) $\{(1, 2), (2, 1)\}$	N	Y	N
(6) $\{(1, 2)\}$	N	N	N

* Ex. 9.1.1

$p = R$

QTP

$\{(a,a), (a,b), (b,a), (b,b), (c,c), (b,c), (c,b)\}$

(1)

reflexive: - Yes

(2)

Symmetric: - Yes

(3)

~~X~~ Transitive: No

$(a,b) \& (b,c) \not\Rightarrow (a,c)$

→ A relation R in $N \times N$ is defined as $(a,b) R (c,d)$ if $a \leq c$ or $b \leq d$. prove that (i) reflexive & (ii) transitive.

Ans (i) $(a,b) R (c,d)$

$(a,b) R (a,b)$ is reflexive iff $a \leq a$ or $b \leq b$ is true & here $a, b, c, d \in N$ and $a \leq c$ or $b \leq d$ is true.

(ii) we can find $(a,b) R (a,b)$ so it is Reflexive

(ii) Transitive $(2,5) R (4,4)$ & $(4,4) R (3,6)$
 $\Rightarrow (2,5) R (3,6)$ here $2 \leq 3$ & $5 \leq 6$ so it is transitive.

* Closure of relation it is transitive.

(i) Reflexive and Symmetric closure:-

(i) $R \cup \Delta_A$ is reflexive closure of R where

$$\Delta_A = \{(a,a) \mid a \in A\}$$

Ex $A = \{a, b, c\}$ & $R = \{(a,a), (a,b), (b,a), (c,b)\}$
 then $(b,b), (c,c) \notin R$

$\therefore S = R \cup \Delta_A$ is reflexive closure where $\Delta_A = \{(a,a), (c,c)\}$

(ii) $R \cup R^{-1}$ is the symmetric closure of R where R^{-1} is the inverse of R .

$$R = \{(a,a), (a,b), (b,a), (c,b)\}$$

$$(b,c) \notin R$$

So, $S = R \cup R^{-1}$ is a symmetric closure where $R^{-1} = \{(b,c)\}$.

(iii) Transitive Closure $\rightarrow$

If R is relation on set A , then the connectivity relation R^* consists of the pair (a,b) such that there is a path between a and b in relation R or

we can say $R^* = \bigcup_{n=1}^{\infty} R^n$

The transitive closure of a relation R is equal to the connectivity relation R^* .

(i) R^* is transitive relation

(ii) S is transitive relation on set A with $R \subseteq S$ then $R^* \subseteq S$

(iv) Partial Ordering:- A relation R on a set A is said to be in Partial Order relation if R is

(i) a reflexive relation (ii) an anti-symmetric (iii) a transitive

(v) Lexicographic Order:- The lexicographic order S on $A_1 \times A_2$ is defined by the dictionary ordering pairs which come first & second. $A_1 = \{a,b\}$ $A_2 = \{a,b\} \Rightarrow R = \{(a,a), (a,b), (b,a), (b,b)\}$

* Language:-

An alphabet can be defined as a finite set of symbols.
we denote alphabet by the set Σ

Ex:

$\Sigma = \{0, 1\}$ for binary sequences,

$\Sigma = \{0, 1, 2, \dots, 9\}$ for decimal no.

$\Sigma = \{a, b, \dots, z\}$ for character string in lower.

$\Sigma = \{A, B, \dots, Z\}$ " " " " Upper

→ ~~Alphabet is known as a symbols in computational field or in TAC or in mathematics.~~

↳ While we combine some symbols with each other by applying some rule we get a new A-string.

or

A string is a finite sequence of symbols selected from some alphabet.

Ex. $\Sigma = \{a, b\}$ alphabet then

String = abab

String

the set of all strings over Σ is denoted by Σ^*

So,
 $\Sigma^* = \{ \Lambda, a, b, aa, ab, ba, bb, aaa, aab, aba, \dots \}$

↳ The length of string x over Σ is the number of symbols in the string & denoted by $|x|$

Ex $|a| = 1, |aa| = 2, |aba| = 3 \dots$ etc

↳ The NULL string is a string over Σ , no matter what alphabet Σ is. It is denoted by Λ &

$$|\Lambda| = 0.$$

↳ So, Now language means a set of strings involving symbols from some alphabets.

Language is a problem of m/c
4 strings are instance of problem

So, A few examples of language over Σ is

$$\{\epsilon, a, aa, aab\}$$

$$\{x \in \{a, b\}^* \mid |x| \leq 8\}$$

$$\{x \in \{a, b\}^* \mid |x| \text{ is odd}\} \text{ etc.}$$

→ For any language over an Alphabet Σ
their union, intersection & difference
are also lang. over Σ

Complement of Language L

$$L' = \Sigma^* - L$$

if

$$L_1 \subseteq \Sigma_1^* \text{ \& } L_2 \subseteq \Sigma_2^*$$

then L_1 & L_2 are subsets of $(\Sigma_1 \cup \Sigma_2)^*$

$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{\epsilon, a, b, aa, ab, ba, bb, aaa, aab, \dots \text{ etc}\}$$

$$L = \{x \in \{a, b\}^* \mid |x| < 3\}$$

$$L = \{\epsilon, a, b, aa, ab, ba, bb\}$$

$$L' = \Sigma^* - L = \{aaa, aab, aba, abb, \dots \text{ etc}\}$$

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⇒ Proof :-

- A Proof of a statement is essentially just a convincing argument that the statement is true but not just convincing but explains why the statement is true.

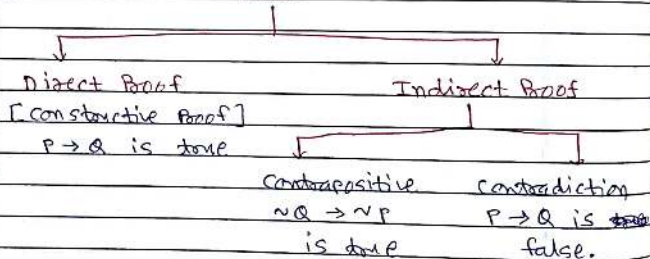
- A typical steps in a Proof is to derive some statement from

(i) assumptions or hypotheses

(ii) statements have already been derived

(iii) other generally accepted facts.

Proof



Date : _____

(i) Direct Proof :- $(P \rightarrow Q)$

- assumes that the statement P is True & uses this to show Q is true

Proof

ex:- for any integer a and b if a and b are odd then a*b is odd.

⇒

P : a and b are odd

Q : a*b is odd

$\exists x \in \mathbb{N} \quad a = 2x+1$ and

$\exists y \in \mathbb{N} \quad b = 2y+1$

$$\begin{aligned} a*b &= (2x+1)*(2y+1) \\ &= (4xy+2x+2y+1) \\ &= 2(\underbrace{2xy+x+y}_M)+1 \end{aligned}$$

let $M = 2xy+x+y$

$$\therefore a*b = 2M+1$$

so, a*b is odd numbers.

Hence Proof :

a and b are odd $\rightarrow$ a*b is also odd.

ex:- Prove, for any integers a and b ,
if a and b are even then $a+b$
is even.

$\Rightarrow$ a if : $a = 2x$
 $b = 2y$

$$\begin{aligned} a+b &= 2x+2y \\ &= 4xy \\ &= 2(2xy) \\ &= 2M \quad \text{where } M = 2xy \end{aligned}$$

$$\therefore a+b = 2M$$

so, $a+b$ is even number.

Hence Prove.

a and b are even $\rightarrow a+b$ is also even

(ii) Contrapositive (Indirect Proof) :-

$\Rightarrow$ $\sim Q \rightarrow \sim P$ is true.

ex: Prove that if $i * j = n$ then
 $i \leq \sqrt{n}$ or $j \leq \sqrt{n}$.

$\Rightarrow$	Note:- $i * j = n$	$i \leq \sqrt{n}$ or $j \leq \sqrt{n}$
	$i = 5$	$5 \leq \sqrt{10}$ $2 \leq \sqrt{10}$
	$j = 2$	$5 \leq 3$ $2 \leq 3$
	$i * j = 10$	false true

Solⁿ:- $P \Rightarrow i * j = n$
 $Q \rightarrow i \leq \sqrt{n}$ or $j \leq \sqrt{n}$

so, $\sim P = i * j \neq n$
 $\sim Q = i > \sqrt{n}$ and $j > \sqrt{n}$

$\sim Q \rightarrow \sim P$ is true.

$\rightarrow i > \sqrt{n}$ [multiplying by j]

$$\therefore i * j > \sqrt{n} * j \quad \text{--- (i)}$$

$\rightarrow j > \sqrt{n}$ [multiplying by $\sqrt{n}$]
 $j * \sqrt{n} > \sqrt{n} * \sqrt{n}$

$$\therefore j * \sqrt{n} > n \quad \text{--- (ii)}$$

Date :

from (i) & (ii)

$$\therefore i * j > \sqrt{n} \quad \text{then} \quad [\because a > b \& b > c, \therefore a > c]$$

$$\therefore i * j \neq n$$

Hence Proved.

(iii) Contradiction (Indirect Proof) :-

$\Rightarrow$ If we have to prove that $P \rightarrow Q$ then the method of contradiction assume that P doesn't imply Q and then try to derive some contradiction.

rule: $P \rightarrow Q$ is false.

Remember :-

$\because P = 1+2=3$ is true then

we try for $1+2 \neq 3$ but in the end we get $1+2=3$ is true

Date :

ex:-

Prove that $\sqrt{2}$ is irrational by method of contradiction.

[GIVEN]

 $\Rightarrow$

$\{x \text{ is a rational if } \exists m, n \in \mathbb{Z}, \text{ such that } x = \frac{m}{n}\}$

$\Rightarrow$ on this defⁿ, let's assume that $\sqrt{2}$ is rational. So,

$\exists m', n'$ such that

$$\sqrt{2} = \frac{m}{n}$$

$\therefore m = \sqrt{2}n$ [squaring both side]

$$\therefore m^2 = 2n^2 \quad \text{--- (i)}$$

$\Rightarrow m^2$ is an even

$\Rightarrow m$ is an even --- (ii)

$\rightarrow \exists K \in \mathbb{Z}$ such that $m = 2K$ --- (iii)

$\Rightarrow$ By (i) & (iii)

$$(2K)^2 = 2n^2$$

$$\therefore 4K^2 = 2n^2$$

common factors $\rightarrow$ rational
no common factors $\rightarrow$ irrational
Date: _____

$$\Rightarrow 2k^2 = n^2$$

$\Rightarrow n^2$ is an even

$\Rightarrow$ n is an even — (iv)

$\Rightarrow$ By (ii) & (iv) m, n are even.

$\Rightarrow$ so, m & n both are divisible by 2.

$\Rightarrow$ so, m & n have common factors.

$\Rightarrow$ A contradiction to statement means that m & n have no common factor.

$\Rightarrow$ so, our assumption ^{is} ~~has~~ wrong.

so $\sqrt{2}$ is irrational.

ex: Prove that $\sqrt{3}$ is irrational.

$\Rightarrow \sqrt{3} = \frac{m}{n}$ • suppose $\sqrt{3}$ is rational num.

$$\therefore m = \sqrt{3}n \quad [\text{squaring both side}]$$

$$\therefore m^2 = 3n^2 \quad \text{--- (i)}$$

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m is divisible by 3. — (ii)

$\Rightarrow \exists k \in \mathbb{Z}$ such that m = 3k — (iii)

$\Rightarrow$ By (i) & (iii)

$$(3k)^2 = 3n^2$$

$$\therefore 9k^2 = 3n^2$$

$$\therefore 3k^2 = n^2$$

$$\therefore n^2 = 3k^2$$

$\therefore$ n is divisible by 3. — (iv)

$\Rightarrow$ By (ii) & (iv) m, n, are divisible by 3.

$\Rightarrow$ so m & n have common factor.

$\Rightarrow$ a contradiction to statement means that m & n have no common factors.

$\Rightarrow$ so our assumption ^{is} ~~has~~ wrong.

so $\sqrt{3}$ is irrational.

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ex:- Prove for any integers a & b if a & b are odd then ab is odd.

$\Rightarrow$ contradiction step: ab is even.

$$\text{so, } a = 2x+1 \text{ (a is odd)}$$

$$b = 2y+1 \text{ (b is odd)}$$

$$a \cdot b = 2z \text{ (ab is even)}$$

$$(2x+1)(2y+1) = 2z$$

$$\therefore 4xy + 2x + 2y + 1 = 2z$$

$$\therefore 2(2xy + x + y) + 1 = 2z$$

$$\therefore \frac{2(2xy + x + y) + 1}{2} = z$$

$$\therefore \frac{2(2xy + x + y)}{2} + \frac{1}{2} = z$$

$$\therefore \frac{(2xy + x + y) + 1}{2} = z$$

$\therefore z$ is not an integer & hence it is contradiction of our prediction.

i.e. a & b are odd then ab is odd.

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ex:- Prove for any sets A, B & C if $A \cap B = \emptyset$ and $C \subseteq B$ then $A \cap C = \emptyset$.

$\Rightarrow$ It can be proved through contradiction.

Let us assume that $A \cap C \neq \emptyset$ & hence we can assume that for some x ,

$$x \in A \cap C$$

$$\text{if } x \in A \cap C \text{ i.e. } x \in A \text{ \& } x \in C$$

└ (i)

$$\text{since } C \subseteq B \text{ \& } x \in C \Rightarrow x \in B$$

└ (ii)

from eqⁿ (i) & (ii) we can say that

$$x \in A, x \in B \text{ and } x \in C$$

$$\text{i.e. } x \in A \cap B$$

But $A \cap B = \emptyset$. Hence there is contradiction on.

so, we proved that

$$\text{if } A \cap B = \emptyset \text{ \& } C \subseteq B \Rightarrow A \cap C = \emptyset$$

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⇒ Principle of Mathematical Induction:-
(PMI)

eg:-

$$\text{Prove that } \sum_{i=0}^n i = \frac{n(n+1)}{2} ; \forall n \geq 0$$

G.T.D

OR

$$\text{Prove that } 1+2+3+\dots+n = \frac{n(n+1)}{2} ; n \geq 0$$

- $P(n)$ is a statement for $n \in \mathbb{N}$ to prove that $P(n)$ is true.

- for every $n \geq n_0$, we show

(i) $P(n_0)$ is true. ← Basic step

(ii) for any $k \geq n_0$, ← Induction hypothesis

if $P(k)$ is true, then $P(k+1)$ is true.

⇒

Step 1: Basic step

for $n=0$ (any no. of n)

We must show that $P(0)$ is true.

$$\text{So, } P(0) = 0 = \frac{0(0+1)}{2}$$

$$0 = 0$$

∴ that is true.

∴ $P(0)$ is true.

eg:-

Note:-

$$\Rightarrow \sum_{i=1}^n 1 = n \quad \sum_{i=1}^n 2 = 2n$$

$$\Rightarrow \sum_{i=1}^n i = \frac{n(n+1)}{2} \text{ series } 1+2+3+\dots+n$$

$$\Rightarrow \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} \text{ series } 1^2+2^2+\dots+n^2$$

$$\Rightarrow \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4} \text{ series } 1^3+2^3+\dots+n^3$$

$$\Rightarrow \prod_{i=1}^n i = 1, 2, 3, \dots, n$$

⇒

Step 2: Induction hypothesis $\sum_{i=1}^n i = n$

$$P(k) = 1+2+3+\dots+k = \frac{k(k+1)}{2} \text{ --- C.D}$$

∴ $P(k)$ is true.

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⇒ Step 3 :- Proof of induction $n=k+1$

Now Prove that $P(k+1)$ is also true

L.H.S :- $\therefore 1+2+3+\dots+k+1 = \frac{(k+1)(k+1+1)}{2}$

$$1+2+3+\dots+k+1$$

$$= 1+2+3+\dots+k+k+1 \quad [\because \text{from (i)}]$$

$$= \frac{k(k+1)}{2} + (k+1)$$

$$= \frac{(k+1) \left[\frac{k}{2} + 1 \right]}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

$$= \frac{(k+1)(k+1+1)}{2}$$

$$= \text{R.H.S}$$

$\therefore P(k+1)$ is done.

$\therefore$ Thus $P(k+1)$ is true, whenever $P(k)$ is true.

→ Hence, from the principle of Mathematical Induction, statement $P(n)$ is true for all natural numbers n .

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ex:- Prove that $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}; n \geq 1$

OR

$$1^2+2^2+3^2+\dots+n^2 = \frac{n(n+1)(2n+1)}{6}; n \geq 1$$

⇒ Step 1: Basic step $n=1$

$$P(n) = 1^2+2^2+3^2+\dots+n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$P(1) = 1^2 = \frac{1(1+1)(2(1)+1)}{6}$$

$$= 1 = \frac{1(2)(3)}{6}$$

$$\therefore 1 = 1$$

$\therefore P(1)$ is true.

⇒ Step 2: Induction hypothesis $n=k$

$$P(k) = 1^2+2^2+3^2+\dots+k^2 = \frac{k(k+1)(2k+1)}{6}$$

$\therefore P(k)$ is true.

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→ step 3: Proof of induction, $n = K+1$

we have;

$$1^2 + 2^2 + 3^2 + \dots + (K+1)^2 = \frac{(K+1)(K+1+1)(2(K+1)+1)}{6}$$

L.H.S = $1^2 + 2^2 + 3^2 + \dots + (K+1)^2$

$$= \frac{1^2 + 2^2 + 3^2 + \dots + K^2 + (K+1)^2}{6}$$

$$= \frac{K(K+1)(2K+1) + (K+1)^2}{6}$$

$$= (K+1) \left[\frac{K(2K+1) + (K+1)}{6} \right]$$

$$= (K+1) \left[\frac{2K^2 + K + 6K + 6}{6} \right]$$

$$= (K+1) \left[\frac{2K^2 + 7K + 6}{6} \right]$$

$$\begin{array}{c} 12 \\ \swarrow \quad \searrow \\ 4 \quad 8 \end{array}$$

$$= (K+1) \left[\frac{2K^2 + 4K + 3K + 6}{6} \right]$$

$$= (K+1) \left[\frac{2K(K+2) + 3(K+2)}{6} \right]$$

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$$= (K+1) \left[\frac{(K+2)(2K+3)}{6} \right]$$

$$= (K+1) \left[\frac{(K+1+1)(2(K+1)+1)}{6} \right]$$

$$= \text{R.H.S}$$

∴ $P(K+1)$ is true.

Hence Proved.

eg:-

Suppose σ is a real number other than 1. Prove that for any $n \geq 0$,

$$\sum_{i=0}^n \sigma^i = \frac{1 - \sigma^{n+1}}{1 - \sigma}$$

⇒

step 1: Basic step

for $n=0$

L.H.S $\sigma^0 = \underline{1}$

R.H.S $\frac{1 - \sigma^{0+1}}{1 - \sigma} = \frac{1 - \sigma}{1 - \sigma} = \underline{1}$

∴ $P(0)$ is true.

∴ $P(n)$ is true.

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⇒ Step 2: Induction step $n=k$

$$1^0 + 1^1 + 1^2 + \dots + 1^k = \frac{1 - 1^{k+1}}{1 - 1}$$

for $k \geq 0$ $P(k)$ is true.

⇒ Step 3: Proof of induction $n=k+1$

$$1^0 + 1^1 + 1^2 + \dots + 1^{k+1} = \frac{1 - 1^{(k+1)+1}}{1 - 1}$$

L.H.S $1^0 + 1^1 + 1^2 + \dots + 1^{k+1}$

$$= 1^0 + 1^1 + 1^2 + \dots + 1^k + 1^{k+1}$$

$$= \frac{1 - 1^{k+1}}{1 - 1} + 1^{k+1}$$

$$= \frac{1 - 1^{k+1} + (1 - 1)(1^{k+1})}{1 - 1}$$

$$= \frac{1 - 1^{k+1} + 1^{k+1} - 1^{k+1}}{1 - 1}$$

$$= \frac{1 - 1^{(k+1)+1}}{1 - 1}$$

= L.H.S

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Ex:

G.T.V

Using PMI Prove that for every $n \geq 1$,

$$1 + 13 + 19 + \dots + (6n+1) = n(3n+4)$$

⇒

Step 1: Basic step

$$\text{for } n=1 \quad 6(1)+1 = 1(3(1)+4)$$

$$\therefore 7=7$$

∴ $P(1)$ is true.

⇒

Step 2: Induction hypothesis $n=k$

$P(k)$ is true

$$1 + 13 + 19 + \dots + (6k+1) = k(3k+4)$$

⇒

Step 3: Proof of Induction $n=k+1$

$$1 + 13 + 19 + \dots + (6(k+1)+1) = (k+1)(3(k+1)+4)$$

L.H.S

L.H.S

$$1 + 13 + 19 + \dots + (6k+1) + (6(k+1)+1)$$

$(k+1) \cdot 6$

$$= k(3k+4) + (6(k+1)+1)$$

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$$= K(3K+4) + 6K+6+1$$

$$= 3K^2 + 4K + 6K + 7$$

$$= 3K^2 + 10K + 7$$

$$= 3K^2 + 3K + 7K + 7$$

$$= 3K(K+1) + 7(K+1)$$

$$= (K+1)(3K+7)$$

$$= (K+1)(3K+3+4)$$

$$= (K+1)(3(K+1)+4)$$

$$= \text{R.H.S}$$

Hence Proved.

ex:- Using PMI Prove that $n^3 + 2n$ is divisible by 3.

$$\Rightarrow \text{Step 1:- } n^3 + 2n = 3m, m \in \mathbb{N}$$

$$\text{L.H.S :- } (1)^3 + 2(1) = 3 \rightarrow \text{It is divisible by 3.}$$

$$\text{R.H.S :- } 3(1) = 3 \rightarrow \text{divisible by 3}$$

$\therefore P(1)$ is true.

$$\Rightarrow \text{Step 2:- } n = k \quad (\text{Induction hypothesis})$$

$$\therefore k^3 + 2k = 3m \quad \text{--- (i)}$$

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$$\Rightarrow \text{Step 3:- Induction Proof } n = k+1.$$

$$(k+1)^3 + 2(k+1) = 3m$$

$$\text{L.H.S } (k+1)^3 + 2(k+1)$$

$$= (k+1)^3 + 2(k+1)$$

$$= (k+1) \cdot (k+1)^2 + 2k+2$$

$$= (k+1)(k^2 + 2k + 1) + 2k + 2$$

$$= k^3 + 2k^2 + k + k^2 + 2k + 1 + 2k + 2$$

$$= k^3 + 3k^2 + 5k + 3$$

$$= k^3 + 3k^2 + 3k + 2k + 3$$

$$= k^3 + 2k + 3k^2 + 3k + 3$$

$$= 3m + 3k^2 + 3k + 3$$

$$= 3[m + k^2 + k + 1], t \in \mathbb{N}$$

$$= 3t \quad \text{divisible by 3.}$$

$\therefore$ R.H.S

$\therefore$ Hence Proved.

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e.g. Prove that any $n \geq 1$, $n(n^2+5)$ is divisible by 6.

$\Rightarrow$ Step 1: Basic step ~~is~~ $n=1$

$P(n=1)$

$$1(1^2+5) = 6 \text{ is divisible by 6.}$$

$\therefore P(1)$ is true.

$\Rightarrow$ Step 2: Induction hypothesis $n=k$

$$k(k^2+5) = 6m \quad \text{--- (i)}$$

$\Rightarrow$ Step 3: Proof of Induction $n=k+1$

$$\text{L.H.S. : } (k+1)(k+1)^2+5$$

$$= (k+1)(k^2+2k+1+5)$$

$$= (k+1)(k^2+2k+6)$$

$$= k^3+2k^2+6k+k^2+2k+6$$

$$= k^3+3k^2+8k+6$$

$$= k^3+3k^2+3k+5k+6$$

$$= k^3+5k+3k^2+3k+6$$

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$$= k(k^2+5) + 3(k^2+k+2)$$

$$= \underset{\substack{\uparrow \\ \text{divisible} \\ \text{by 6}}}{6m} + 3 \underset{\substack{\uparrow \\ \text{divisible by 6}}}{(k(k+1)+2)}$$

$\Rightarrow$ Now, consider $3(k(k+1)+2)$

$k(k+1)$ is product of two consecutive no. as one of them must be even.

$\Rightarrow k(k+1)$ is multiple of 2. Then,

$(k(k+1)+2)$ is also multiple of 2.

$\therefore 3(k(k+1)+2)$ is multiple of 6.

$\Rightarrow \therefore (k+1)(k+1)^2+5$ is divisible by 6.

$\therefore n(n^2+5)$ is divisible by 6.

Hence Proved.

e.g. $5^n - 2^n$ is divisible by 3, $n \geq 1$

$\Rightarrow$ Step 1 $n=1$

$$5^1 - 2^1 = 3 \rightarrow \text{divisible by 3}$$

$\therefore P(1)$ is true.

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⇒ Step 2: Induction hypothesis $n=k$.

$$5^k - 2^k = 3m \quad \text{--- (i)}$$

∴ $P(k)$ is true.

⇒ Step 3: Proof of Induction $n=k+1$

$$\begin{aligned} \text{L.H.S.} &= 5^{k+1} - 2^{k+1} \\ &= 5^k \cdot 5 - 2^k \cdot 2 \\ &= \cancel{5^k} \cdot \cancel{5} + 3 \cdot 5^k - \cancel{2^k} \cdot \cancel{2} \\ &= 5^k \cdot 5 - 2^k \cdot 5 + 2^k \cdot 3 \\ &= 5(5^k - 2^k) + 3 \cdot 2^k \\ &= 5(3m) + 3 \cdot 2^k \\ &\quad \uparrow \quad \quad \uparrow \\ &\quad \text{divisible} \quad \text{divisible by 3} \\ &\quad \text{by 3} \end{aligned}$$

∴ $5(3m) + 3 \cdot 2^k$ is divisible by 3

∴ $5^{k+1} - 2^{k+1}$ is divisible by 3.

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eg. By PMI Prove that $(10)^{2n} - 1$ is divisible by 11. $n \geq 1$

⇒ Step 1: $n=1$ Basic step

$$\begin{aligned} &(10)^{2(1)} - 1 \\ &= 100 - 1 \\ &= 99 \rightarrow \text{divisible by 11.} \end{aligned}$$

∴ $P(1)$ is true.

⇒ Step 2: Induction hypothesis $n=k$

$$(10)^{2k} - 1 = 11m \quad \text{--- (i)}$$

⇒ Step 3: Proof of Induction $n=k+1$

$$\begin{aligned} &= (10)^{2(k+1)} - 1 \\ &= 10^{2k+2} - 1 \\ &= (10)^{2k} \cdot (10)^2 - 1 \\ &= 100 \cdot (10)^{2k} - 100 + 99 \\ &= 100((10)^{2k} - 1) + 99 \\ &= 100(11m) + 99 \quad \text{divisible by 11.} \end{aligned}$$

eg. Prove that $\sum_{i=1}^n i^3 = \frac{n^2 (n+1)^2}{4}$

⇒ ① Basic step $n=1$

$$P(n) = 1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2 (n+1)^2}{4}$$

$$\begin{aligned} \text{L.H.S} &= n^3 \\ &= 1 \\ \text{R.H.S} &= \frac{1(1+1)^2}{4} \\ &= \frac{1}{1} \end{aligned}$$

∴ $P(1)$ is true.

② Induction hypothesis $n=k$

$$P(k) = 1^3 + 2^3 + 3^3 + \dots + k^3 = \frac{k^2 (k+1)^2}{4} \quad \text{--- (i)}$$

③ Proof of Induction $n=k+1$

$$\text{Let } 1^3 + 2^3 + 3^3 + \dots + (k+1)^3 = \frac{(k+1)^2 (k+2)^2}{4}$$

$$\begin{aligned} \text{L.H.S} &= 1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 \\ &= \frac{k^2 (k+1)^2}{4} + (k+1)^3 \end{aligned}$$

$$= \frac{(k+1)^2}{4} [k^2 + 4(k+1)]$$

$$= (k+1)^2 \left[\frac{k^2 + 4k + 4}{4} \right]$$

$$= (k+1)^2 (k+2)^2$$

$$= \frac{(k+1)^2 (k+2)^2}{4} = \text{R.H.S}$$

$$= \frac{(k+1)^2 (k+2)^2}{4}$$

∴ Hence Proved.

Note:-

$$\sum_{i=1}^n 2^i = 2 + 2^2 + 2^3 + 2^4 + \dots + 2^n$$

$$\sum_{i=1}^n i^2 = 1^2 + 2^2 + \dots + n^2$$

$$\sum_{i=0}^n 2^i = 1 + 2 + 2^2 + 2^3 + \dots + 2^n$$

$$\sum_{i=0}^n i^2 = \sum_{i=1}^n i^2$$

$$\sum_{i=0}^n 2^i \neq \sum_{i=1}^n 2^i$$

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eg. Using P.M.T. Prove that if $0 < a < 1$ then $(1-a) \cdot n \geq 1-na$, $n \geq 1$

$\Rightarrow$ (1) Basic step :- $n=1$ and $0 < a < 1$

$$(1-a) \cdot 1 \geq 1-(1)a$$

$$\therefore 1-a \geq 1-a$$

$\therefore P(1)$ is true.

(2) Induction Hypothesis $n=k$

$$P(k) = (1-a) \cdot k \geq 1-ka$$

(3) Proof of Induction hypothesis $n=k+1$

$$(1-a) \cdot (k+1) \geq 1-(k+1) \cdot a$$

$$\text{L.H.S } (1-a) \cdot (k+1)$$

$$= k+1-ak-a$$

$$= k-ka+1-a$$

$$= k \cdot (1-a) + 1-a$$

$$\geq 1-ka+1-a$$

$$\geq 2-a(k+1)$$

$$\geq 1-(a \cdot (k+1))$$

$\therefore$ Hence $P(k+1)$ is true.

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eg. Using P.M.T. Prove that

$$1+3+5+\dots+\alpha = n^2 \text{ for all } n \geq 0$$

where α is an odd integer & n is the number of terms in the sum. ($\alpha = 2n-1$)

$\Rightarrow$ Basic step $n=1$

for $n \geq 0$

$$\alpha = 2n-1 = 2-1 = 1 \text{ \& } n^2 = 1^2 = 1.$$

$n \geq 0$, $1+3+5+\dots+\alpha = n^2$ is true.

$\therefore P(1)$ is true.

$\Rightarrow$ Induction Hypothesis :-

for $n=k$

$$\therefore 1+3+5+\dots+\alpha(2k-1) = k^2$$

where $\alpha = 2k-1$

$\Rightarrow$ Proof of Induction $n=k+1$

$$\therefore 1+3+5+\dots+\alpha(2(k+1)-1) = (k+1)^2$$

$$\therefore 1+3+5+\dots+\alpha(2k+1) = (k+1)^2$$

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$$\text{L.H.S} \therefore 1+3+5+\dots+(2K-1)+(2K-1+2)$$

$$\therefore 1+3+5+\dots+(2K-1)+(2K+1)$$

$$\therefore K^2 + 2K + 1$$

$$\therefore (K+1)^2$$

$\therefore$ R.H.S

$\therefore P(K+1)$ is also true.

ex- show that $2^n > n^3$ for $n > 10$ by M.I

$\Rightarrow$ ① Basic step $n=11$

$$\text{RHS } 2^{11} = 2048$$

$$\text{LHS } 11^3 = 1331$$

$$\therefore 2^{11} > 11^3$$

$\therefore P(11)$ is true.

② Induction hypothesis $n=k$

$$\therefore P(K) = 2^K > K^3 ; K > 10$$

③ Proof of Induction $n=k+1$

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$$\text{ie } 2^{K+1} > (K+1)^3$$

$$\text{L.H.S} = 2^{K+1}$$

$$= 2^K \cdot 2$$

$$> 2^K \cdot \left(1 + \frac{1}{10}\right)^3$$

$$> 2^K \cdot \left(1 + \frac{1}{K}\right)^3 \quad \left(\because K > 10 \text{ but } \frac{1}{K} < \frac{1}{10}\right)$$

$$> K^3 \cdot \frac{(K+1)^3}{K^3} \quad (\because \text{step 2})$$

$$> (K+1)^3$$

Hence $P(K+1)$ is true.

ex-
=

define M.I.P & show that for any $n \geq 4$,

$$n! > 2^n$$

$\Rightarrow$ ① Basic step for $n=4$

$$\text{L.H.S } n! = 4!$$

$$= 4 \times 3 \times 2 \times 1$$

$$= 24$$

$$\text{R.H.S } 2^n = 2^4$$

$$= 16$$

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$$\therefore 24 > 16$$

$$\therefore n! > 2^{k+1}$$

$\therefore P(4)$ is done.

(ii) (2) Induction Hypothesis $n=k$

$$\therefore k! > 2^k$$

(3) Proof of Induction $n=k+1$

$$\text{i.e. } (k+1)! \geq 2^{k+1}$$

$$\begin{aligned} \text{L.H.S } (k+1)! &= (k+1) \cdot k! \\ &\geq 2 \cdot k! \quad \left[\because k \geq 4, k+1=5 \right] \\ &\geq 2 \cdot 2^k \quad \left[\because 5 \geq 2 \right] \\ &\geq 2^{k+1} \end{aligned}$$

$$\therefore (k+1)! \geq 2^{k+1}$$

Hence $P(k+1)$ is done.

$$\left(\frac{n(n+1)}{2} \right)^2$$

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P.V.

$$\sum_{i=0}^n i^3 = \left(\sum_{i=0}^n i \right)^2 = \frac{n^2(n+1)^2}{4}$$

STEP: 3

$$\text{L.H.S } = 1 + 8 + 27 + \dots + (k+1)^3 \quad \left[\because \sum_{i=0}^n i = \frac{n(n+1)}{2} \right]$$

$$= 1 + 8 + \dots + k^3 + (k+1)^3$$

$$= \frac{k^2(k+1)^2}{4} + (k+1)^3$$

$$= \frac{(k+1)^2}{4} (k^2 + (k+1)4)$$

$$= \frac{(k+1)^2}{4} (k^2 + 4k + 4)$$

$$= \frac{(k+1)^2}{4} (k+2)^2$$

$$= \frac{(k+1)^2 (k+1+1)^2}{4}$$

$$= \text{R.H.S} \quad \therefore \text{Hence } P(k+1) \text{ is done.}$$

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eg. show by M.I. that $n^4 - 4n^2$ is divisible by 3, for all $n > 0$.

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$$\text{eg. } \sum_{i=1}^n \frac{1}{i(i+1)} = \frac{n}{n+1}$$

$\Rightarrow$ ① Basic step $n=1$

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

$$\text{L.H.S. } \frac{1}{1(1+1)} = \frac{1}{2} \quad \text{R.H.S.} = \frac{1}{(1+1)} = \frac{1}{2}$$

$\therefore P(1)$ is true.

② Induction Hypothesis $n=k$

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1} \quad \text{--- (1)}$$

③ Proof of induction $n=k+1$

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{(k+1)(k+2)} = \frac{k+1}{(k+1)+1}$$

$$\text{L.H.S. } \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)}$$

$$= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)}$$

$$= \frac{k(k+2) + 1}{(k+1)(k+2)}$$

$$= \frac{k^2 + 2k + 1}{(k+1)(k+2)} \Rightarrow \frac{(k+1)^2}{(k+1)(k+2)+1}$$

$$\Rightarrow \frac{k+1}{(k+1)+1}$$

$\therefore P(k+1)$ true.

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⇒ Language :-

- An alphabet can be defined as a finite set of symbols.

- We denote alphabet by the set Σ .

eg. $\Sigma = \{0, 1\}$ for binary sequence.

$\Sigma = \{0, 1, 2, \dots, 9\}$ for decimal no.

$\Sigma = \{a, b, \dots, z\}$ for character string in lower.

$\Sigma = \{A, B, \dots, Z\}$ for character string in upper.

↳ While we combine some symbols with each other by applying some rule we get a new string.

ex: $\Sigma = \{a, b\}$ alphabet then

string = abab

the set of all strings over Σ is denoted by Σ^* .

$\Sigma^* = \{\epsilon, a, b, aa, ab, ba, aaa, aab, aba, \dots\}$

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↳ The length of string x over Σ is the number of symbols in the string & denoted by $|x|$.

eg. $|a| = 1, |aa| = 2, |aba| = 3, \dots$

↳ The NULL string is a string over Σ . No matter what alphabet Σ is. It is denoted by ϵ .

$|\epsilon| = 0$

defⁿ Language :-

✓ A set of strings involving symbols from some alphabets.

↳ ✓ Language is a problem of machine & string are instance of problem.

eg: few examples of language over Σ is

$\{\epsilon, a, aa, aab\}$

$\{x \in \{a, b\}^* \mid |x| \leq 8\}$

$\{x \in \{a, b\}^* \mid |x| \text{ is odd}\}$ etc..

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→ For any language over an alphabet Σ , their union, intersection & difference are also language over Σ .

⇒ Complement of language L ,

$$L' = \Sigma^* - L$$

if $L_1 \subseteq \Sigma_1^*$ & $L_2 \subseteq \Sigma_2^*$

then L_1 & L_2 are subsets of $(\Sigma_1 \cup \Sigma_2)^*$

$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{\Lambda, a, b, aa, ab, ba, bb, aaa, aab, \dots\}$$

$$L = \{x \in \{a, b\}^* \mid |x| < 3\}$$

$$L = \{\Lambda, a, b, aa, ab, ba, bb\}$$

$$L' = \Sigma^* - L$$

$$= \{aaa, aab, aba, abb, \dots\}$$

⇒ Concatenation operation:-

- If x & y are elements of Σ^* the concatenation of x & y is the

$$L_1 \cdot L_2 = \{xy \mid x \in L_1, y \in L_2\}$$

$$L \{\Lambda\} = \{\Lambda\} = L$$

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String xy formed by writing the symbols of x and symbol y consecutively.

$$\text{If } x = abb \text{ \& } y = ba$$

$$xy = abbb a \text{ \& } yx = baabb$$

$$\Rightarrow x \Lambda = \Lambda x = x$$

$$\Rightarrow (xy)z = x(yz) \text{ (Associative)}$$

⇒ Substring:-

- x is a substring of another string y , if there are string w & z , either or both of which may be null.

So that,

$$y = wxz$$

eg:-

Suppose $x = car$ & $y = descarter$

x is a substring of y .

> A Prefix of a string is an initial substring.

- A suffix of a string is a final substring.

✓- Normally Prefix of a string is Λ .

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$$\Rightarrow L^* = \bigcup_{i=0}^{\infty} L^i \quad * = \text{Kleen star}$$

= strings are concatenation of zero or more symbols.

= Λ is always element of L^* .

$$\Rightarrow L^+ = \bigcup_{i=1}^{\infty} L^i$$

$$= L \cdot L^*$$

$$= L^* \cdot L$$

eg. show that for any language L

$$L^* = (L^+)^* = (L^+)^+ = (L^*)^+$$

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$\Rightarrow$ Recursive definition of a language:-

- A language over an alphabet Σ can be described recursively.

- A recursive definition has 3 steps:

- ① Specify some basic objects in the set.
- ② specify the rules for constructing more objects from the objects already known.
- ③ Declaration that no objects except those constructed as given above are allowed in the set.

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ex:- Define a lang. of Polynomials recursively.
give derivation for $5x^2 + 7x$.

⇒ We should be able to derive a term.

A polynomial can be derived

Rule 1: Any no. is term

2: variable x is term.

3: P & Q are term $\Rightarrow PQ$ is also term.

4: A term is polynomial.

5: If P & Q are polynomial $\Rightarrow P+Q$ & $P-Q$ is also term.

Deriving $5x^2 + 7x$

Rule

5 is a term 1

x is a term 2

$5x$ is a term 3

$5x \cdot x$ is a term 3

$5x^2$ is polynomial 4

7 is a term 1

x is a term 2

$7x$ is a term 3

$5x^2 + 7x$ is a polynomial 5

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ex:- Give the recursive definition of PAL. of Palindrome over any alphabet Σ .

⇒ for a string to be palindrome, first and the last element of the string must be same recursively.

→ Let Σ be any alphabet. The lang. PAL of Palindrome over Σ can be defined as follows.

Rule 1 :- $\Lambda \in \text{PAL}$

2 :- for any $a \in \Sigma$, $a \in \text{PAL}$

3 :- for any $x \in \text{PAL}$, & any $a \in \Sigma$ $axa \in \text{PAL}$

4: No string is in PAL unless it can be obtained by using rules 1, 2 & 3.

Note:- just for note:-

$\Sigma = \{a, b\}$

Rule 1: a & b are in PAL.

Rule 2: If x is in PAL, then axa & bxb

But all the words in the language defined above have an odd length so its not correct.

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→ If λ_{eq} represents a saturation of

eg. Recursive definition of L^* .

⇒ Rule 1: $\wedge \in L^*$

2: HGL^* & YGL^* , $XYGL^*$

3. No string is in L^* unless it
"can't" be obtained by using
Rule 1 + Rule 2.

eg. set of Arithmetic Expression.

$$\Rightarrow \Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, +, -, *, /, ()\}$$

rules:-

2) Any number (positive, negative or zero) is in A.E.

2) If x is in A , then so are (x) and $-(x)$.

3) If x and y are in AE , then

(i) $x+y$ (if the 1st symbol in y is not)

(ii) $x-y$ (" " " " " "

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$$k \in \mathbb{N} \quad x \neq y$$

cin 3/4

(V) $x \neq y$

(4) AE consists of only those things can be created by the above rules.

$(5 * (8 + 2)) \leftarrow \text{generated}$

$$5 - (8+1)/3 \leftarrow$$

$(6+7)/9$ and $4(19*4) \leftarrow$ not generated

$$\Rightarrow 8 * 6 - ((4/2) + (3-1) + 7) 4 \rightarrow$$

1) Each of the numbers are in AE by Rule 1

2) $8 * 6$ is in AE by rule 3 (iii)

3) $4/2$ is in AE by rule 3 (iv).

4) $C(4,2)$ is in AE by Rule 2

5) $3-1$ is in AE by rule 3Cii)

6) $C3-1$ is in AE by Rule 2

7) $(3-1)*7$ is in AE by rule 3 (iii)

8) $(4|2) + (3|-1) * 7$ is in AE by rule 3ci).

8) $(C4|2) + (3-1)*7$ is in AE by rule ~~3~~ 2

10) $(C(4|2) + (3-1)*7)/4$ is in AE by rule 3 (iv)

11) $8 * 6 + ((4/2) + (3-1) * 7) / 4$ is in AE by rule 3(i).

eg.

The set T of positive integer divisible by 2 or 7.

 $\Rightarrow$

$$2 \in T, 7 \in T$$

for $\forall n \in T$ & every positive integer i
 $i * n \in T$

eg.

The set V of all string in $\{0,1\}^+$ containing the substring 00.

 $\Rightarrow$

$$00 \in V$$

for $\forall x \in V$, all strings $0x$, $1x$ and $x0$, $x1$ are in V .

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⇒ Strong Principle of mathematical Induction

- Suppose $P(n)$ is statement involving an integer n . Then to prove that $P(n)$ is true for every $n \geq n_0$, it is sufficient to show these:

- 1) $P(n_0)$ is true.
- 2) for any $k \geq n_0$, $P(n)$ is true for every n satisfying $n_0 \leq n \leq k$, then $P(k+1)$ is true.

eg:- $f(n) < \left(\frac{5}{3}\right)^n$ for $\forall n \geq 0$.

⇒ Basic step:

$$f(0) = 1$$

$$\therefore f(0) < \left(\frac{5}{3}\right)^0 \text{ is true.}$$

⇒ Induction hypothesis.

$k \geq 0$, for $\forall n$ $0 \leq n \leq k$,

$$f(n) < \left(\frac{5}{3}\right)^n$$

⇒ statement to show in induction step:

$$f(k+1) < \left(\frac{5}{3}\right)^{k+1}$$

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⇒ Proof of Induction step:-

$$\begin{aligned} f(k+1) &= f(k) + f(k-1) \\ &\leq \left(\frac{5}{3}\right)^k + \left(\frac{5}{3}\right)^{k-1} \\ &\leq \left(\frac{5}{3}\right)^{k-1} \left(\frac{5}{3} + 1\right) \\ &\leq \left(\frac{5}{3}\right)^{k-1} \left(\frac{8}{3}\right) \\ &\leq \left(\frac{5}{3}\right)^{k-1} \left(\frac{24}{9}\right) \\ &\leq \left(\frac{5}{3}\right)^{k-1} \left(\frac{25}{9}\right) \\ &\leq \left(\frac{5}{3}\right)^{k-1} \left(\frac{5}{3}\right)^2 \\ &\leq \left(\frac{5}{3}\right)^{k+1} \end{aligned}$$

Hence Proved.

GTV

Prove that statement $(P \vee Q) \rightarrow R$ and $(P \rightarrow R) \vee (Q \rightarrow R)$ are logically equivalent.

$\Rightarrow$

P	Q	R	$P \vee Q$	(i) $(P \vee Q) \rightarrow R$	(ii) $(P \rightarrow R)$	(iii) $(Q \rightarrow R)$	(iv) $(P \rightarrow R) \vee (Q \rightarrow R)$
1	1	1	1	1	1	1	1
1	1	0	1	0	0	0	0
1	0	1	1	1	1	1	1
1	0	0	1	0	0	1	1
0	1	1	1	1	1	1	1
0	1	0	1	0	1	0	1
0	0	1	0	1	1	1	1
0	0	0	0	1	1	1	1

Hence ;

$(P \vee Q) \rightarrow R \Rightarrow (P \rightarrow R) \vee (Q \rightarrow R)$ logical equivalent.

$\Rightarrow$ Note:- results of both are not same
= so it is not logically equivalent

logical equivalent $(\Rightarrow)$

<u>P</u>	<u>Q</u>
True	True
false	True/false
True	True

logical equivalence $(\Leftarrow)$

<u>P</u>	<u>Q</u>
True	True
false	false
True	True